📊 Data Analyst Project Report

Project Title: Healthcare Dashboard for Performance & Financial Analysis
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1. Executive Summary

This project simulates a corporate-grade, end-to-end data analytics workflow for healthcare organizations. The goal was to consolidate fragmented datasets into a unified analytics platform, enabling leadership to monitor patient demographics, track admissions, analyze revenue, and evaluate insurance provider performance.

The final deliverable is an interactive **Power BI dashboard** supported by SQL queries, Power Query transformations, and DAX measures, designed to provide actionable insights for operational efficiency and improved patient outcomes.

2. Business Problem Statement

Healthcare organizations face challenges due to fragmented data across clinical, financial, and patient record systems. This lack of integration hinders the ability to:

- Identify bottlenecks in revenue and admission processes.
- Analyze patient demographics (age, blood type, gender) in relation to medical conditions.
- Monitor resource utilization (doctors, hospitals) and treatment effectiveness.
- Track insurance provider performance and reimbursement trends.

Impact if unresolved: inefficient resource allocation, poor visibility into patient outcomes, delayed decision-making, and missed opportunities for cost optimization.

3. Methodology

Stage	Tools Used	Key Activities
Data Loading	Power Query Imported raw dataset	initial inspection
Exploratory Data Analysis (EDA)	Power BI, SQL	Identified missing values, outliers, distribution patterns
Data Cleaning	Power Query	Removed duplicates, handled nulls, standardized formats
SQL Queries	PostgreSQL	Created tables, ran joins, aggregations, trend analysis
Dashboard Creation	Power BI + DAX	Built KPIs, charts, slicers; calculated YoY, MoM growth
Reporting	Power BI, PPT	Summarized findings, prepared recruiter-friendly presentation

4. Dashboard Features

- **Revenue Analysis:** Total revenue trends and YoY admissions.
- **Admissions Overview:** Patient inflow and discharge patterns.
- **Resource Utilization:** Number of hospitals and doctors.
- **Patient Demographics:** Distribution by blood type, age buckets, gender.
- **Medical Conditions:** Breakdown of diagnoses and treatments.
- **Insurance Providers:** Performance and reimbursement trends.
- **Medication Usage:** Insights into prescription patterns.
- **Dynamic Slicers:** Filters by doctor, hospital, insurance provider, admission date.

5. Key Insights

- **Revenue Trends:** Admissions drive revenue growth, but reimbursement delays impact cash flow.
- **Patient Demographics:** Majority of admissions fall within 30–50 age bucket; blood type O most common.
- **Insurance Providers:** Provider A shows faster reimbursements, Provider C lags significantly.
- **Resource Utilization:** Certain hospitals are overburdened, while others remain underutilized.
- **Medical Conditions:** Chronic conditions (diabetes, hypertension) dominate admissions, requiring preventive strategies.

6. Business Recommendations

- **Optimize resource allocation** by redistributing patient loads across hospitals.
- **Negotiate with underperforming insurance providers** to improve reimbursement timelines.
- **Invest in preventive care programs** targeting chronic conditions to reduce admissions.
- **Enhance staffing strategies** to balance doctor workloads and improve patient outcomes.
- **Leverage demographic insights** to tailor healthcare services for dominant age groups.

7. Deliverables

- **Dataset:** `healthcare_dataset.csv`
- **SQL Scripts:** Database creation, queries, aggregations.
- **Power BI Dashboard:** Interactive visuals with DAX measures.
- **Report & Presentation:** Structured findings and recommendations.

8. Repository & Usage

git clone https://github.com/Kaditya9/Healthcare_Dashboard_data_analysis_SQL-PowerQuery-PowerBI.git cd Healthcare_Dashboard_data_analysis_SQL-PowerQuery-PowerBI

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- Open dataset: `healthcare_dataset.csv`
- Load into PostgreSQL SQL Server
- Connect SQL Database to Power BI
- Apply Power Query transformations
- Use DAX measures (`DAX_Used_For_Comprehensive Healthcare Performance & Financial Analysis`)
- View dashboard: `Comprehensive Healthcare Performance & Financial Analysis.pdf`

9. License

MIT — free to use, fork, and share.

10. About the Author

👨‍💻 **K Aditya** — Data Analyst passionate about breaking down complex data topics into practical, job-ready projects.